

- 4 Meristematic tissue is found in the growing regions of plants, such as shoot tips. Meristematic cells have a similar role to stem cells in animals.

Fig. 4.1 shows some of the stages in the formation of a mature phloem sieve tube element and companion cells from a meristematic cell.

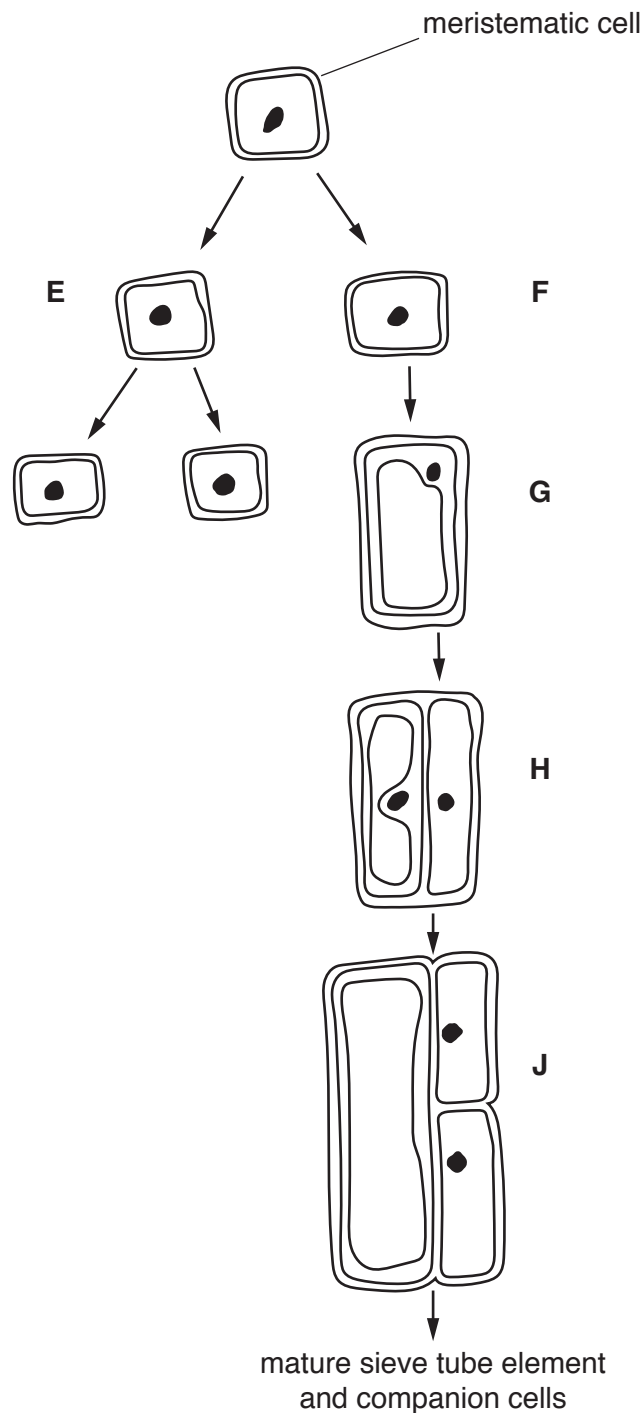


Fig. 4.1

- Explain why it is important that one of the daughter cells (cell **E**) is a meristematic cell.

(b) Complete Table 4.1 to describe the changes that are shown in Fig. 4.1 between stages:

- ### Table 4.1

[3]

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- This image shows a blank sheet of white paper with ten horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

(d) Describe the functions of companion cells in transport in the phloem.

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[Total: 10]